

Evaluate each geometric series described.

11) $-3 + 15 - 75 + 375 \dots, n = 8$

12) $2 + 8 + 32 + 128 \dots, n = 8$

13) $a_1 = 1, r = 4, n = 7$

14) $a_1 = 3, r = 2, n = 7$

15) $\sum_{k=1}^8 -2 \cdot 6^{k-1}$

16) $\sum_{m=1}^8 32 \cdot \left(\frac{1}{2}\right)^{m-1}$

17) $\sum_{i=1}^{10} 0.2 \cdot 5^{i-1}$

18) $\sum_{n=1}^{10} -2 \cdot 2^{n-1}$

Determine the number of terms n in each geometric series.

19) $\sum_{i=1}^n -4^{i-1} = -341$

20) $a_1 = -1, r = -5, S_n = 104$

Determine if each geometric series converges or diverges.

21) $-1 + 2 - 4 + 8 \dots$

22) $-16 - 4 - 1 - \frac{1}{4} \dots$

23) $\sum_{k=1}^{\infty} -3 \cdot \left(\frac{2}{5}\right)^{k-1}$

24) $\sum_{i=1}^{\infty} 2 \cdot 2^{i-1}$

Evaluate each infinite geometric series described.

25) $\sum_{i=1}^{\infty} \left(\frac{1}{3}\right)^{i-1}$

26) $\sum_{i=1}^{\infty} 0.4 \cdot 0.9^{i-1}$

27) $\sum_{m=1}^{\infty} \left(-\frac{2}{3}\right)^{m-1}$

28) $\sum_{k=1}^{\infty} -4^{k-1}$